Introduction to Mathematics for Deep Learning: Midterm Solutions

1. **Mix of Questions** *(50 points, 10 points each)*

1) Let $Z_1, Z_2, Z_3, \ldots$ be i.i.d. random variables that take values "0" and "1" with equal probability. Further, let

$$X_i = \sum_{j=1}^{i} Z_j, \text{ for } 1 \leq i \leq n. \tag{1}$$

Find $I(X_1; X_2, X_3, \ldots, X_n)$.

2) Let X, Y, Z be three binary Bernoulli($\frac{1}{2}$) random variables that are pairwise independent; that is, $I(X;Y) = I(X;Z) = I(Y;Z) = 0$.

i. Under this constraint, what is the minimum value for $H(X,Y,Z)$ (under all possible joint distributions of (X,Y,Z) that are consistent with the above)?

ii. Give an example achieving this minimum.

iii. Now suppose that X, Y, Z are three random variables each uniformly distributed over the alphabet $\{1, 2, \ldots, m\}$. Again, they are pairwise independent. What is the minimum value for $H(X,Y,Z)$?

3) Find the mutual information $I(X;Y)$, where

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim N_2 \left(0, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right), \quad -1 < \rho < 1.$$

Evaluate $I(X;Y)$ for $\rho = 0$, and the limit of $I(X;Y)$ as ρ approaches 1. Comment on your findings.

4) Let $X \sim \mathcal{N}(0,1)$ and $Y|X \sim \mathcal{N}(X, X^2)$. Find the minimum mean squared estimate(MMSE) of Y given X. I.e., find $\hat{Y}_{MMSE}(X)$. What is the corresponding mean squared error?

5) Suppose M is $d \times d$ positive semidefinite matrix. Show that $P^T M P$ is positive semidefinite matrix for any $d \times d$ matrix P.

Solution:

1)

$$\begin{aligned} I(X_1; X_2, X_3, \ldots, X_n) =& H(X_2, X_3, \ldots, X_n) - H(X_2, X_3, \ldots, X_n | X_1) \\ =& H(X_2) + \sum_{i=3}^{n} H(X_i | X_2, \ldots, X_{i-1}) - \sum_{i=2}^{n} H(X_i | X_1, \ldots, X_{i-1}) \end{aligned}$$

$$
\begin{aligned}
&= H(X_2) + \sum_{i=3}^{n} H(Z_i) - \sum_{i=2}^{n} H(Z_i) \\
&= H(X_2) - H(Z_2) \\
&= \frac{3}{2} - 1 \\
&= \frac{1}{2},
\end{aligned}
$$

since X_2 takes value "0" with probability $1/4$, value "1" with probability $1/2$ and value "2" with probability $1/4$, which gives $H(X_2) = 3/2$.

2) i. Due to pairwise independence and the marginal distributions, $H(X, Y) = H(X) + H(Y) = 2$. Thus,

$$
\begin{aligned}
H(X, Y, Z) &= H(X, Y) + H(Z|X, Y) \\
&= H(X) + H(Y) + H(Z|X, Y) \\
&\geq H(X) + H(Y) \\
&= 2
\end{aligned}
$$

with equality if and only if Z is a deterministic function of X and Y. This minimum is achieved by the example in part (b).

ii. Let $Z = X \oplus Y$, where $\oplus$ denotes XOR. It is easy to check that all the marginal distributions are satisfied, as well as pairwise independence.

iii. Here is one possible solution. Without loss of generality, relabel the alphabet to be $\{0, 1, 2, \ldots, m-1\}$ instead of $\{1, 2, \ldots, m\}$. Let $Z = X + Y \mod m$. Then

$$
H(X, Y, Z) = H(X) + H(Y) = \log m + \log m = 2 \log m.
$$

One can then justify that under this construction, all the marginal distributions are satisfied, and we also have pairwise independence. For example, to argue pairwise independence between Z and X,

$$
\begin{aligned}
I(X; Z) &= H(Z) - H(Z|X) \\
&= \log m - H(X + Y|X) \\
&= \log m - H(Y|X) \\
&= \log m - H(Y) \\
&= \log m - \log m \\
&= 0.
\end{aligned}
$$

Thus, X and Z are independent.

3)

$$
\begin{bmatrix} X \\ Y \end{bmatrix} \sim \mathcal{N}_2 \left(\mathbf{0}, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right) \tag{2}
$$

Using the expression for the entropy of a multivariate normal

$$h(X,Y) = \frac{1}{2}\log(2\pi e)^2|K| = \frac{1}{2}\log(2\pi e)^2\sigma^4(1-\rho^2). \tag{3}$$

Since X and Y are individually normal with variance σ^2,

$$h(X) = h(Y) = \frac{1}{2}\log 2\pi e\sigma^2. \tag{4}$$

Hence

$$I(X;Y) = h(X) + h(Y) - h(X,Y) = -\frac{1}{2}\log(1-\rho^2). \tag{5}$$

i. $\rho = 0$. In this case, X and Y are independent, and hence $I(X;Y) = 0$, which agrees with the formula.

ii. It is easy to see that $\lim_{\rho\to 1} -\frac{1}{2}\log(1-\rho^2) = \infty$. This is reasonable since $\rho = 1$ implies $X = Y$, and knowing X implies perfect knowledge about Y. Hence the mutual information should be infinite, which agrees with the formula.

4)

$$\hat{Y}_{MMSE} = \mathbb{E}[Y|X] = X. \tag{6}$$

The error is

$$\begin{aligned}
\mathbb{E}\left[(\hat{Y}_{MMSE}(X) - X)^2\right] =& \mathbb{E}[(Y-X)^2] && (7)\\
=& \mathbb{E}[\mathbb{E}[(Y-X)^2|X]] && (8)\\
=& \mathbb{E}[X^2] && (9)\\
=& 1. && (10)
\end{aligned}$$

5) For any v, we have

$$v^T P^T M P v = (Pv)^T M (Pv) \geq 0. \tag{11}$$

Thus, $P^T M P$ is positive semidefinite.

2. **The prime number theorem** *(25 points)*

Some time around 300 B.C., someone showed that there are infinitely many prime numbers – we know this because a proof appears in Euclid's famous *Elements*. In this problem, we will not only show that there are infinitely many prime numbers, but we will also give a lower bound on the rate of their growth using information theory.

Let $\pi(m)$ denote the number of primes no greater than m. Note that every positive integer m has a **unique** prime factorization of the form

$$m = \prod_{i=1}^{\pi(m)} p_i^{X_i},$$

where $p_1, p_2, p_3, \ldots$ are the primes, that is, $p_1 = 2$, $p_2 = 3$, $p_3 = 5$, etc., and $X_i = X_i(m)$ is the non-negative integer representing the multiplicity of p_i in the prime factorization of m.

Let n be an integer, and N be uniformly distributed random variable on $\{1, 2, 3 \ldots n\}$.

(a) *(10 points)* Show that $X_i(N)$ is an integer-valued random variable satisfying

$$0 \leq X_i(N) \leq \log n. \tag{12}$$

[Hint : Try finding a lower and an upper bound for $p_i^{X_i(N)}$]

(b) *(15 points)* Show that

$$\log n = H(N) \leq \pi(n) \log(\log n + 1). \tag{13}$$

Thus, not only is $\pi(n) \to \infty$ but in fact $\pi(n) \geq \frac{\log n}{\log(\log n+1)}$.

[Hint : Do $X_1(N), X_2(N), \ldots, X_{\pi(n)}(N)$ determine N? What does that say about the respective entropies?].

Solution:

(a) $0 \leq X_i(N)$ is trivial. Note also that $2^{X_i} \leq p_i^{X_i} \leq N \leq n$.
Thus, combining both results, $0 \leq X_i(N) \leq \log n$, as we wanted to show.

(b)

$$\begin{aligned}
\log n &= H(N) && (14)\\
&= H(X_1, X_2 \ldots X_{\pi(n)}) && (15)\\
&= \sum_{i=1}^{\pi(n)} H(X_i | X_1, \ldots, X_{i-1}) && (16)\\
&\leq H(X_1) + H(X_2) + \ldots H(X_{\pi(n)}) && (17)\\
&\leq \pi(n) \log(\log n + 1), && (18)
\end{aligned}$$

where the first step follows because there is a one-to-one mapping between N and $X_1, X_2 \ldots X_{\pi(n)}$. The second step is by the chain rule for entropy. The next step is because conditioning reduces entropy, and the last one is because the distribution that maximizes entropy is the uniform one, there are $\pi(n)$ entropy terms, and the X_i's can take at most $\log n + 1$ different values.

3. **The Mixture of Experts** *(25 points)*

Let the results of an election be the random variable X, where

$$X = \begin{cases} +1 & \text{if candidate A wins} \\ -1 & \text{if candidate B wins} \end{cases} \tag{19}$$

The prior probability of candidate A winning is $p_X(1) = \alpha$. Now m experts try to predict the outcome of X with various degrees of accuracy, perhaps determined by their past performance. The i-th expert predicts $Y_i = Z_i X$, where

$$Z_i = \begin{cases} +1 & \text{with probability } p_i \\ -1 & \text{with probability } 1 - p_i \end{cases} \tag{20}$$

We assume that $Z_1, Z_2, \ldots, Z_m$, and X are all independent. Thus, the prediction Y_i can be viewed as a noisy measurement of X, where the probability of correct prediction is p_i. We are trying to answer the question: "What is the best way to combine these predictions?"

(a) *(5 points)* According to the MAP rule, we pick $\hat{X} = +1$ if

$$p_{X|Y^m}(+1|y^m) > p_{X|Y^m}(-1|y^m). \tag{21}$$

Derive an equivalent inequality in terms of the probability mass function (pmf) of Z^m. I.e., rewrite the above inequality using p_{Z^m}, y^m and α.

(b) *(10 points)* Show that the MAP rule $\hat{X}(y_1, \ldots, y_m)$ is to decide

$$\hat{X} = \begin{cases} +1 & \text{if } \sum_{i=1}^m w_i y_i \geq t \\ -1 & \text{otherwise} \end{cases} \tag{22}$$

for some weights w_i's and threshold t.
Hint: a function $g(y)$ takes input $+1$ or -1 where $g(+1) = p$ and $g(-1) = 1 - p$. Then, we can express the function $g(y) = p^{(1+y)/2}(1-p)^{(1-y)/2}$.

(c) *(5 points)* Find the weights $w_1, \ldots, w_m$ and theshold t. Note that the weights do not depend on the prior probability α.

(d) *(5 points)* Suppose $p_1 = \cdots = p_m > 1/2$. Show that MLE (ignoring prior probability α) is simply majority vote.

Solution:

(a) MAP rule compares the probability of X given $Y_1, \ldots, Y_m$. The MAP rule returns $\hat{X} = +1$ if

$$p_{Y^m|X}(y^m|+1)p_X(+1) > p_{Y^m|X}(y^m|-1)p_X(-1) \tag{23}$$

which implies

$$Pr(Z_1 = y_1, \ldots, Z_m = y_m) \times \alpha > Pr(Z_1 = -y_1, \ldots, Z_m = -y_m) \times (1 - \alpha). \tag{24}$$

I.e.,

$$p_{Z^m}(y^m)\alpha > p_{Z^m}(-y^m)(1 - \alpha). \tag{25}$$

where $-y^m = (-y_1, \ldots, -y_m)$.

(b) Using the above hint, we have

$$\prod_{i=1}^{m} p_i^{(1+y_i)/2}(1 - p_i)^{(1-y_i)/2} \times \alpha > \prod_{i=1}^{m} p_i^{(1-y_i)/2}(1 - p_i)^{(1+y_i)/2} \times (1 - \alpha). \tag{26}$$

Taking log, then

$$\begin{aligned} &\sum_{i=1}^{m} \left[\frac{1 + y_i}{2} \log p_i + \frac{1 - y_i}{2} \log(1 - p_i) \right] + \log \alpha \\ &> \sum_{i=1}^{m} \left[\frac{1 - y_i}{2} \log p_i + \frac{1 + y_i}{2} \log(1 - p_i) \right] + \log(1 - \alpha) \end{aligned} \tag{27}$$

Finally, we have

$$\sum_{i=1}^{m} \left[y_i \log \frac{p_i}{1 - p_i} \right] > \log \frac{1 - \alpha}{\alpha} \tag{28}$$

(c) $w_i = \log \frac{p_i}{1-p_i}$ and $t = \log \frac{1-\alpha}{\alpha}$

(d) MLE is simply equivalent to MAP with $\alpha = 1/2$. If all p_i's are the same and larger than $1/2$, then all w_i's are the same and positive Thus, the decision rule of MLE is simply $\sum_{i=1}^{m} y_i \geq 0$, which is basically majority vote.